Akhil Mohammad

Integrated B.E. in Computer Science and M.Sc. in Economics, 2027

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Education

Birla Institute of Technology and Science, Pilani – Hyderabad Campus *2022 – 2027*
CGPA: 7.59/10, Integrated B.E. in Computer Science and M.Sc. in Economics

Experience

SDE Intern, Porter *Jul 2026 – Dec 2026*

- Developed **REST APIs** within a **Kotlin/Ktor** backend integrated with a **Pub/Sub** messaging layer and **SQL** persistence, supporting production features and experiments carried out from PRD to deployment.
- On **on-call rotation**, handling operational triage across **Datadog** (monitors, APM/traces, Logs Explorer) alongside **ArgoCD**, **Grafana**, **Snowflake**, and **GCS** for deployment and infra monitoring

Defence Research and Development Organisation (DRDO), Hyderabad *May 2025 – Jul 2025*

- Built a **SAR-based ship detection** system integrating deep learning inference with AIS metadata for identity-aware maritime surveillance, with interactive analysis and visualization via a **Streamlit GUI**.
- Engineered custom preprocessing, annotation reconstruction, and validation pipelines to convert heterogeneous SAR datasets, including OpenSAR and SSDD, into YOLO-ready detection data; trained and evaluated YOLO-based models, achieving **strong baseline localization performance on SSDD (mAP@0.5 = 0.875)**.

National Centre for Polar and Ocean Research, Goa *May 2024 – Jul 2024*

- Developed and evaluated time-series forecasting models **(CNNs, RNNs, GRUs, and LSTMs)** on univariate weather data with **TensorFlow and Keras**.
- **Designed and fine-tuned 1D-CNN models** to extract temporal features from sequential weather observations collected at Bharati Station, Antarctica.
- Implemented a **Temporal Convolutional Network** and achieved a Normalized RMSE score of 0.074.

Anukta Infotech Pvt. Ltd. *May 2025 – Jun 2025*

- Designed a **feedback-driven** RAG pipeline with document chunking, semantic retrieval, LLM-based response generation, and feedback-conditioned knowledge updates via **FAISS** re-embedding
- Built a **natural-language-to-SQL system** converting informal queries into executable SQL against structured databases, with results surfaced via a Streamlit interface

Projects

Distributed KV Store with Tunable Consistency – Distributed Systems *Mar 2026*

- Designed a **distributed key-value store** in Python with leader-follower replication, implementing version-based conflict resolution to handle concurrent writes across 3 nodes
- Built **dual-mode read semantics**, eventual (local read) and strong (quorum read across nodes), exposing consistency as a **tuneable client parameter** via REST API
- Implemented **heartbeat-based failure detection** and automatic leader re-election using a deterministic selection rule, enabling the cluster to recover without manual intervention

Hindi Information Retrieval Engine – Search Systems *Feb 2026*

- Built an end-to-end IR system with a positional **inverted index** over a 600K+ token Hindi corpus, enabling efficient Boolean and phrase queries
- Optimized retrieval using **gap encoding** and **variable-byte compression** (3.5× reduction) and implemented **tolerant search with k-gram indexing** and transliteration for wildcard and cross-script queries

Skills

- **Backend:** REST APIs, Kotlin, Ktor, Flask, Pub/Sub, SQL, Postgres
- **Distributed Systems & Databases:** Replication, Consistency Models, Inverted Index, Query Processing
- **Cloud & Infra:** ArgoCD, Kubernetes, Docker, Terraform, GCS
- **Observability & Data:** Datadog, Grafana, Snowflake
- **AI & ML:** Time-series, Computer Vision, TensorFlow, PyTorch, RAG, Information Retrieval, Langchain
- **Programming Languages:** Python, C++, Java, C, R, JavaScript, Kotlin